Lab: Stored XSS into HTML context with nothing encoded

Platform: PortSwigger
Category: Cross-Site Scripting (XSS)
Difficulty: Apprentice
Tools: Browser

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1. Objective

The goal of this lab is to exploit a Stored XSS vulnerability in the comment functionality by submitting a comment that executes JavaScript when the blog post is viewed.

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2. Impact

An attacker can inject JavaScript into a comment and store it on the website. When a user views the affected blog post, the injected JavaScript can execute in their browser.

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3. Steps to Solve

Step 1: Find the comment functionality

I opened the lab and browsed the blog post.
I found a comment section where users can submit comments. Since the comment is displayed on the blog page after being submitted, I decided to test whether the application properly handles HTML input.

Step 2: Inject the XSS payload

I entered the following payload into the comment field:

- `<script>alert(1)</script>`

The `<script>` tag is used to execute JavaScript, while `alert(1)` displays a simple alert box.
I also entered the required name, email, and website fields.

Step 3: Submit the comment

I clicked **Post comment** to submit the comment.
The application accepted the comment and stored it on the blog post.

Step 4: View the blog post

I went back to the blog post and viewed the page again.
The stored payload was loaded with the comment and executed by the browser.
A JavaScript alert appeared showing: `1`
This confirmed that the Stored XSS was successfully exploited

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4. Payload

The payload used was: `<script>alert(1)</script>`

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5. Result

The JavaScript executed successfully when the blog post was viewed.
The alert box displayed `1`, confirming the Stored XSS vulnerability.

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6. Root Cause

The vulnerability exists because the application stores the comment and later renders it in the HTML page without properly encoding the user input.
As a result, the browser interprets the injected `<script>` tag as executable JavaScript instead of normal text.

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7. Conclusions and Evaluation

What did I learn?

I learned how Stored XSS works when user input is saved by the application and executed later when the affected page is viewed. I also learned how to identify a simple HTML injection point and confirm it using a JavaScript `alert()`.

Lab Rating

4/5 ⭐ ⭐ ⭐ ⭐

Did I face any challenges?

The lab was straightforward. The main thing I needed to understand was that the payload was stored with the comment and executed when the blog post was viewed again.

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8. References

- PortSwigger Web Security Academy - Cross-Site Scripting
- PortSwigger Web Security Academy - Stored XSS
- PortSwigger Web Security Academy - Stored XSS into HTML context

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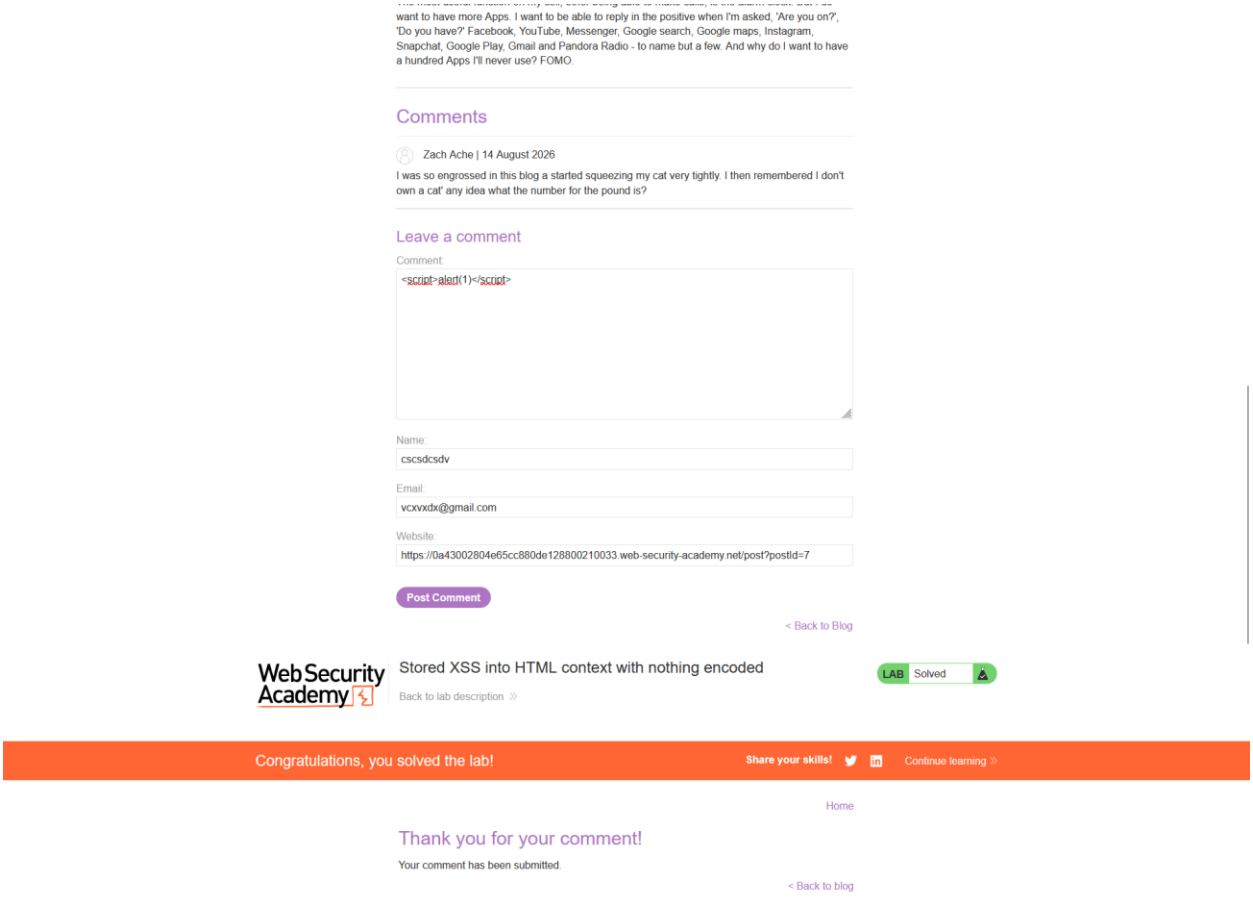
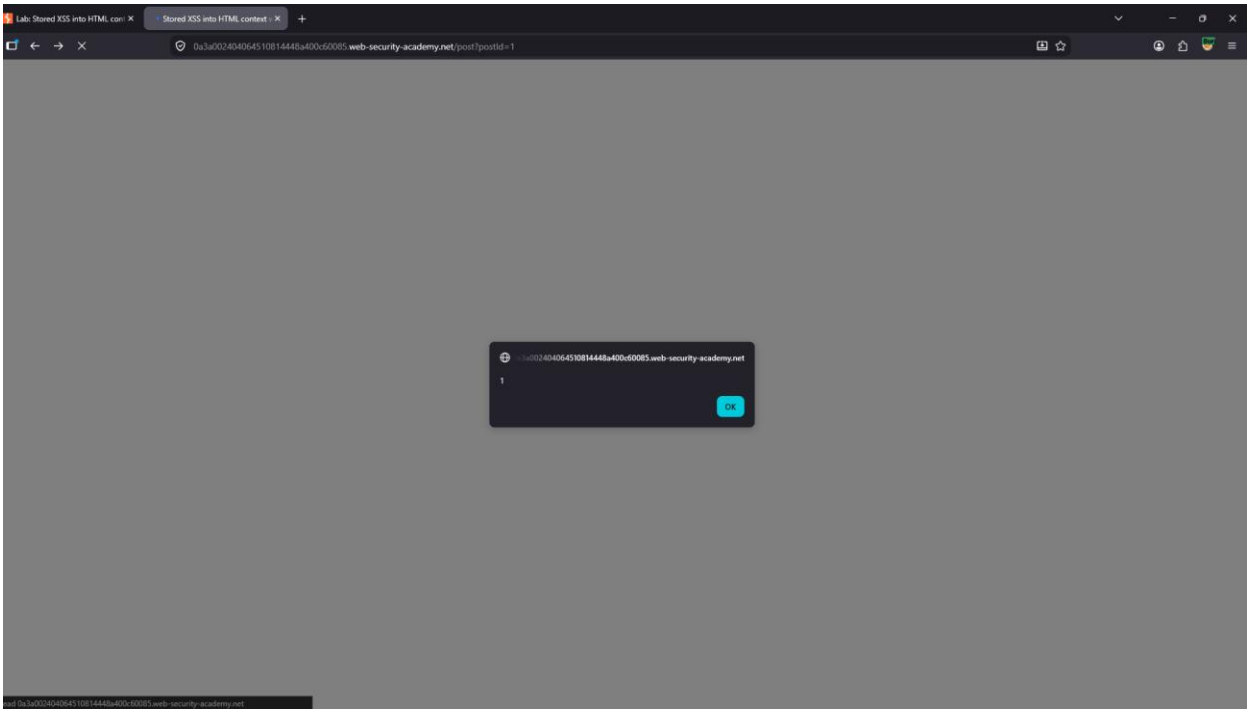
8. Screenshots

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